Python Programming Foundations

Week 6 - API's, External Data, and Responsible AI Use

Day 3: "Python Beyond Console Scripts"

1

Python Can Live Beyond One Script

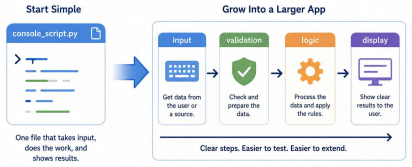
Python can run as a console script, but it can also be part of a larger application.

The ideas are familiar. The responsibilities are separated more clearly.

You are not starting over. You are seeing how input, logic, data, and display can be organized when a program grows.

2

From Console Script to Larger App

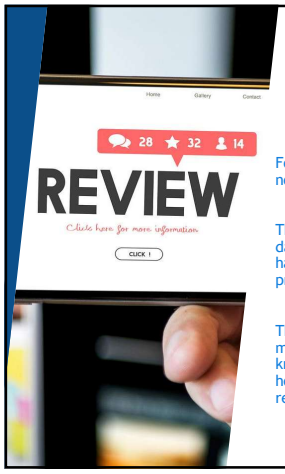


Small beginnings build strong foundations for bigger programs.

Python Can Live Beyond One Script

3

Review: API Output Still Needs Explanation



For Assignment 11, the goal is not just to retrieve data.

The goal is to explain where the data came from, what shape it had, and which values your program selected.

That same explanation habit matters in larger applications: know where information enters, how it is handled, and what result is shown.

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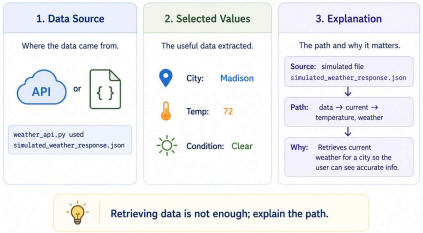
A11 Closeout



Retrieving data is not enough; explain the path.

Review: API Output Still Needs Explanation

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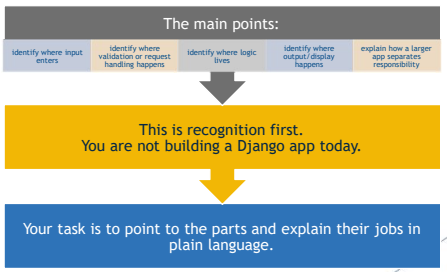
What Counts as Success Today



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Recognize Larger App Architecture
Same ideas. More structure. Clear responsibilities.
Small Console Script
One file handles everything. Great place to start.
All responsibilities live in one file.
Recognition Path (5 Steps)
1 Input
Get data from the user or a source.
2 Validation
Check and prepare the data.
3 Logic
Process the data and apply rules.
4 Output / Display
Show clear results to the user.
5 Separated Responsibilities
Each part has its own job and place.
Larger App Flow
Responsibilities live in separate parts.
Recognition first. No Django build today.
Name where responsibilities live so you can grow with confidence.
Today's Success

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What We Will Use Today
input
validation
logic
display
template
view
model as recognition vocabulary
These words help us talk about responsibility. They are not a requirement to build a full framework project.

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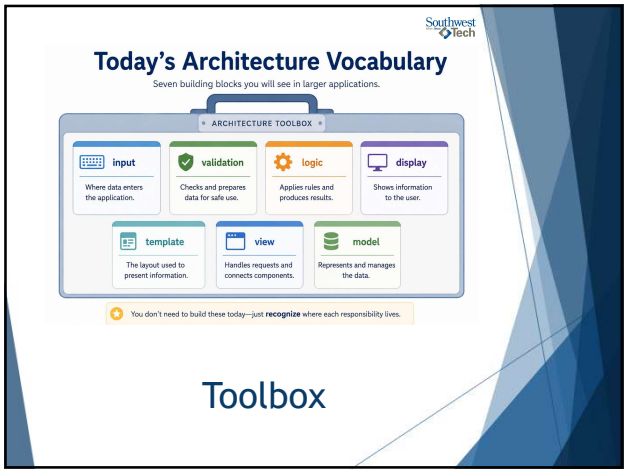
Today's Architecture Vocabulary
Seven building blocks you will see in larger applications.
ARCHITECTURE TOOLBOX
input
Where data enters the application.
validation
Checks and prepares data for safe use.
logic
Applies rules and produces results.
display
Shows information to the user.
template
The layout used to present information.
view
Handles requests and connects components.
model
Represents and manages the data.
You don't need to build these today—just recognize where each responsibility lives.
Toolbox

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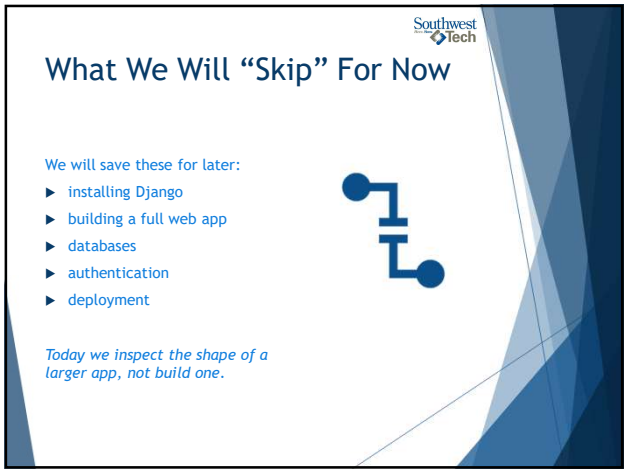
What We Will "Skip" For Now
We will save these for later:
installing Django
building a full web app
databases
authentication
deployment
Today we inspect the shape of a larger app, not build one.

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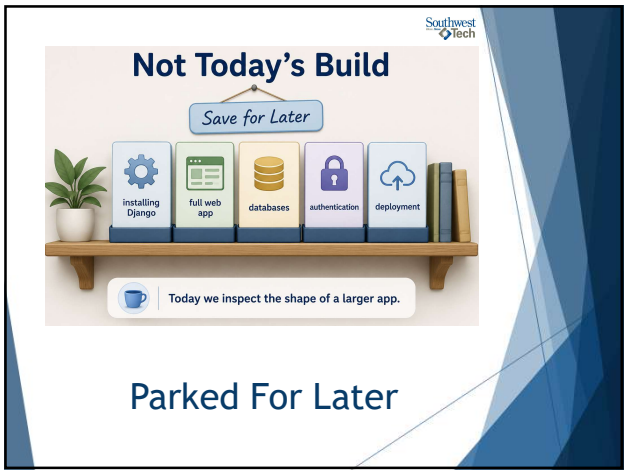
Not Today's Build
Save for Later
installing Django
full web app
databases
authentication
deployment
Today we inspect the shape of a larger app.
Parked For Later

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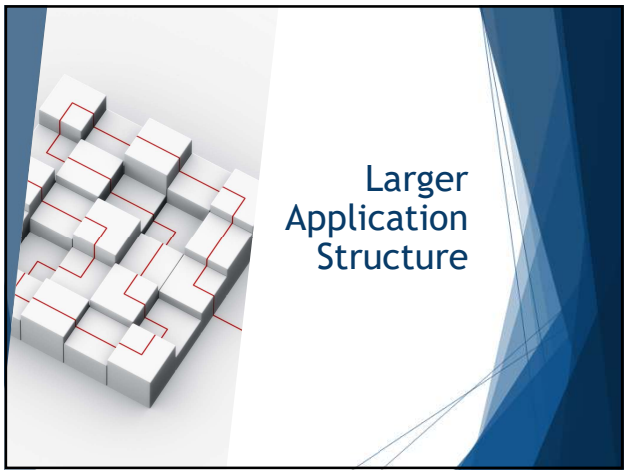
Larger Application Structure

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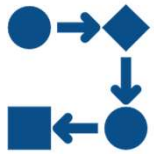
Larger Apps Separate Responsibilities



As programs grow, it becomes useful to separate responsibilities.

Different parts of the application can handle input, validation, data logic, and display.

Separation makes the app easier to read, change, test, and explain because each part has a clearer job.

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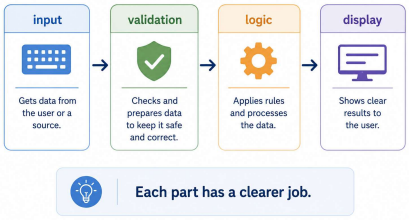
Larger Apps Separate Responsibilities



Each part has a clearer job.

Larger Apps Separate Responsibilitie

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Input Enters Through A Controlled Place

In a console script, input may come from `input()`.

In a larger app, input may come through a form, request, or controlled interface.

The source changes, but the question stays familiar: what information is coming in, and what should the program do with it?

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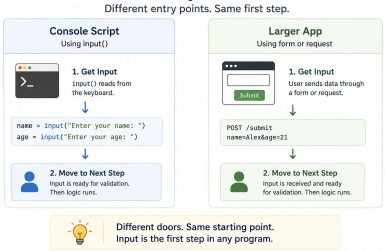
Where Input Enters

Different entry points. Same first step.



Different doors. Same starting point. Input is the first step in any program.

Input Enters Through A Controlled Place

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Validation Protects The Flow

Validation checks whether input is acceptable before the program depends on it.

This is the same idea as checking file data or API response data before using it.

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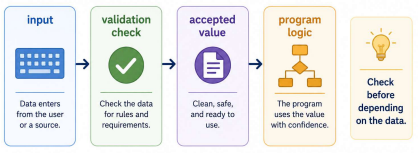
Validation Protects The Flow

Good data in, good results out.



Validate early. Trust results later.

Validation Protects The Flow

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MVT As Recognition Vocabulary

MVT is one way to talk about separated responsibilities:

- Model: data shape and rules
- View: request handling and decisions
- Template: displayed result

Today you only need to recognize the flow.

You may also hear the term MVC. The names shift across frameworks, but the shared idea is the same: separate data, logic, and display.

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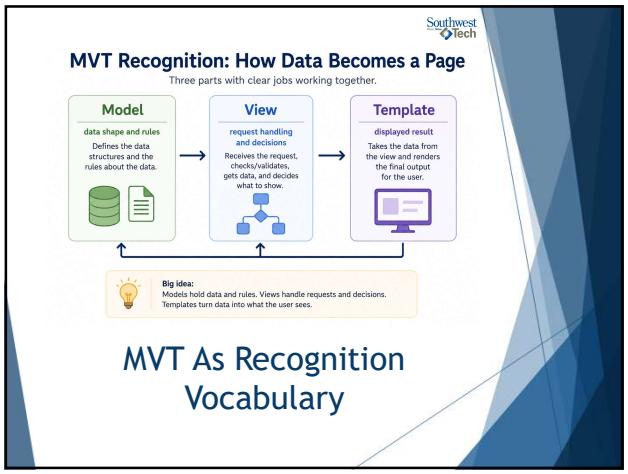

MVT As Recognition Vocabulary

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MVC And MVT Use Different Names

You may also hear MVC: Model, View, Controller.

Django uses MVT: Model, View, Template.

The names are not identical:

- MVC View is closer to what the user sees.
- Django Template is what the user sees.
- Django View handles request decisions.

The shared idea is responsibility separation.

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Full Stack As A Preview

A full-stack application connects multiple parts:

- what the user sees
- the logic that handles requests
- the data the app stores or retrieves
- the flow between those parts

In this course, you only need to recognize the idea. You are *not expected to build a full-stack app*.

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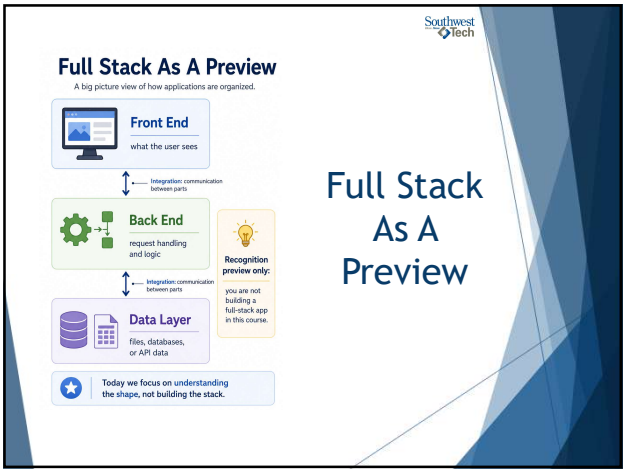

Full Stack As A Preview

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Console Flow Versus App Flow

A console script may run from top to bottom.

A larger app often responds to a request, validates input, runs logic, and then returns a display.

That means the flow may be request-driven instead of simply line-by-line from the top of one file.

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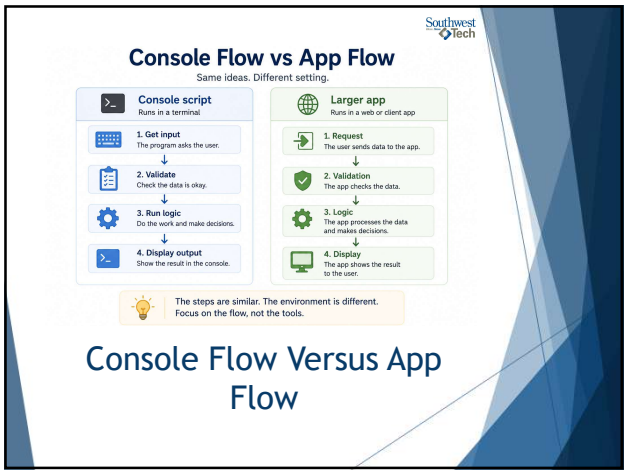

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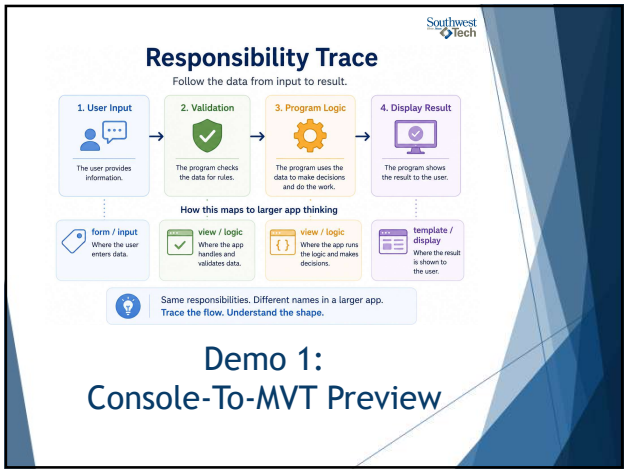

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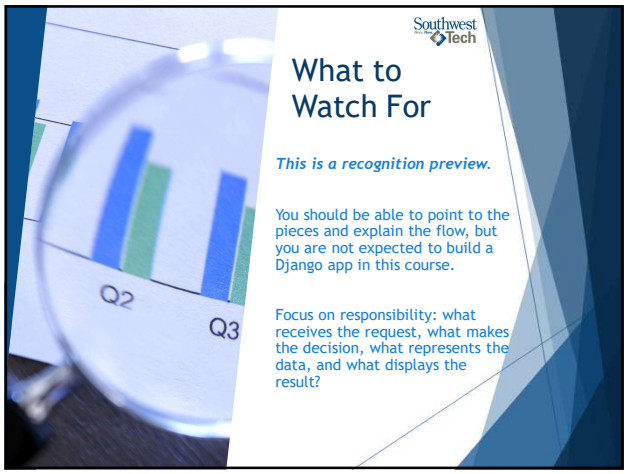

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MVT Recognition Preview

Three parts. Clear jobs. Working together.

MODEL — data shape
Defines the data structures and rules.

VIEW — request handling
Receives the request, processes data, and decides what to do.

TEMPLATE — display
Turns the result into output for the user.

Recognize the flow; do not build Django today.

Demo 2: Django MVT Recognition

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Common Failure: Preview Becomes Panic

Seeing a larger structure does not mean you are behind.

Recognition comes before implementation. Today, naming the parts and explaining the flow is the success target.

This preview is meant to make later courses less surprising, not to create a new hidden requirement.

DON'T Panic

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Preview Is Not Panic

You are building understanding, not the whole system.

Recognition today
Your goal is to see the shape.
- Learn the parts and their jobs.
- See how they connect.
- Understand the flow.

Focus: awareness and clarity.

Implementation later
Your goal is to build with confidence.
- Apply the parts in a real project.
- Make good design choices.
- Iterate and improve over time.

Focus: practice and growth.

Naming the parts is the success target.
Today: recognize. Later: implement.

Common Failure: Preview Becomes Panic

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Assignment #12

For Assignment 12: Inspect a guided Python app-architecture example.

Identify:
- where input enters
- where validation happens
- where logic lives
- where output is displayed

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A12 Architecture Preview

Map the flow at a high level. Keep it recognition-only.

1. **Where does input enter?** — Name the place where user data comes in.
2. **Where does validation happen?** — Name the step or place that checks the data.
3. **Where does logic live?** — Name where the program does the work and makes decisions.
4. **Where is output displayed?** — Name where the user sees the result.

Goal: Identify the purpose of each part and how they connect. Recognition today. Implementation later.

Assignment 12

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AI Use In Lab 0n

- AI can suggest API code that looks believable but does not match the actual response.
- You must inspect the response, run the code, and explain what human decision you made.
- If AI names a field that does not exist in your response, your program is still wrong even if the code looks professional.

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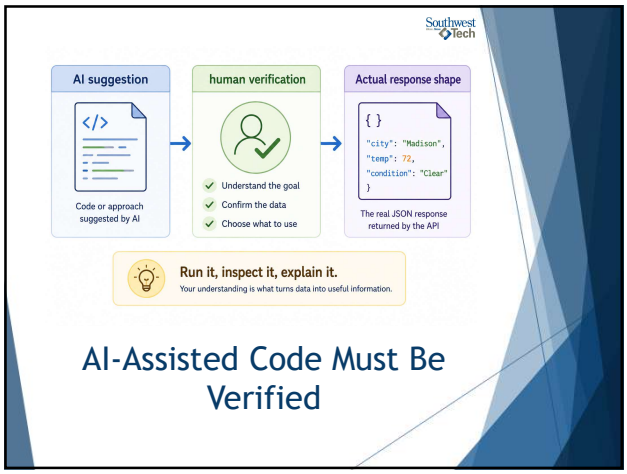
AI suggestion
human verification
Actual response shape
Code or approach suggested by AI
Understand the goal
Confirm the data
Choose what to use
{ }
"city": "Madison",
"temp": 72,
"condition": "Clear"
}
The real JSON response returned by the API
Run it, inspect it, explain it.
Your understanding is what turns data into useful information.
AI-Assisted Code Must Be Verified

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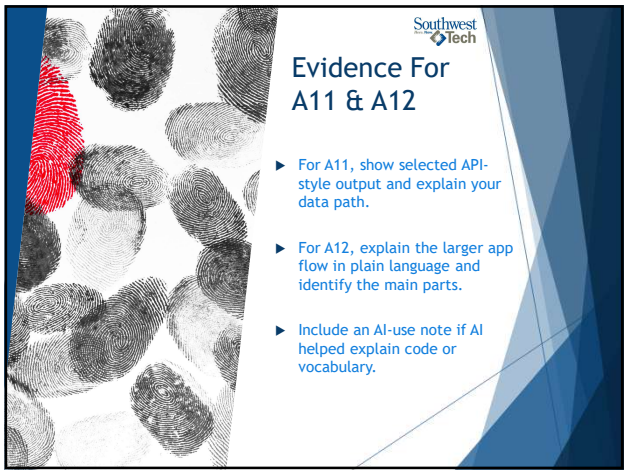
Evidence For A11 & A12
For A11, show selected API-style output and explain your data path.
For A12, explain the larger app flow in plain language and identify the main parts.
Include an AI-use note if AI helped explain code or vocabulary.

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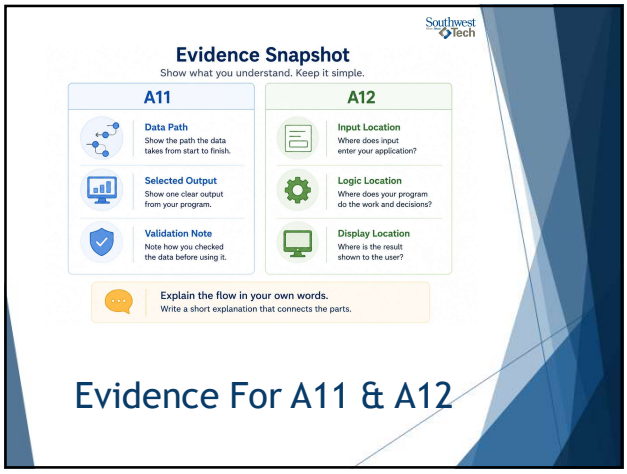
Evidence Snapshot
Show what you understand. Keep it simple.
A11
Data Path
Show the path the data takes from start to finish.
Selected Output
Show one clear output from your program.
Validation Note
Note how you checked the data before using it.
A12
Input Location
Where does input enter your application?
Logic Location
Where does your program do the work and decisions?
Display Location
Where is the result shown to the user?
Explain the flow in your own words.
Write a short explanation that connects the parts.
Evidence For A11 & A12

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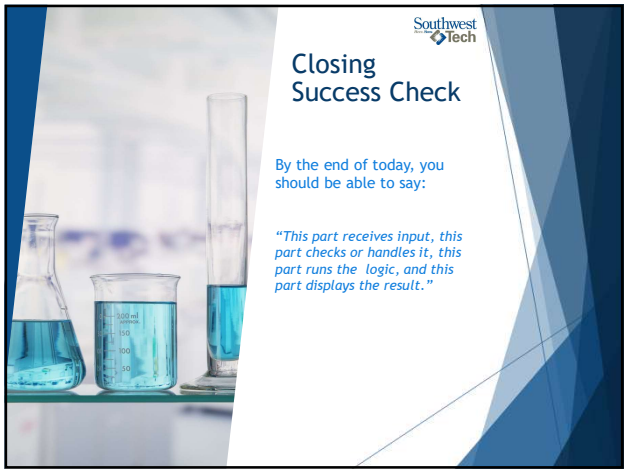
Closing Success Check
By the end of today, you should be able to say:
"This part receives input, this part checks or handles it, this part runs the logic, and this part displays the result."

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Thank you!
Have Fun!

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